

CSE230: Discrete Mathematics

Practice Sheet 1: Propositional Logic Basics

Q1	Let p, q and r be the propositions "The student has submitted the assignment", "The professor has graded the assignment", "The student receives a passing grade" respectively. Express each of these compound propositions as an English sentence. $p \wedge \neg q$ $p \vee r$ $p \rightarrow q \wedge r$ $p \leftrightarrow q$ $\neg r \wedge (\neg p \vee \neg q)$
Q2	Let p, q, r and s be the propositions p: The student has submitted the research paper. q: The student has attended all workshops. r: The student has participated in group projects. s: The student will receive a degree with honors. Express each of these compound propositions as an English sentence. $p \vee (q \wedge \neg s)$ $\neg p \vee (q \wedge r)$ $(p \wedge r) \rightarrow s$ $(p \wedge q) \leftrightarrow r$ $(p \rightarrow q) \wedge (r \rightarrow s)$ $\neg(p \wedge s) \vee (q \wedge r)$
Q3	Let p and q be the propositions: p: The weather is sunny. q: We will go to the beach. Write these propositions using p and q and logical connectives (including negations and exclusive OR): - It is sunny and we go to the beach. - It is sunny but we do not go to the beach. - It is not sunny and we do not go to the beach. - Either it is sunny, or we go to the beach (or both). - If it is sunny, then we go to the beach. - Either it is sunny or we go to the beach, but not both, and if it is sunny, then we do not go to the beach - That it is sunny is necessary and sufficient for going to the beach.
Q4	Let p, q, and r be the propositions: p: The campground is open for visitors. q: Campfires are allowed in the campground. r: The river nearby is safe for swimming. Write these propositions using p, q, and r and logical connectives (including negations). - The campground is open for visitors, but campfires are not allowed. - Campfires are allowed, and the river is safe for swimming, but the campground is not open for visitors. - If the campground is open, then campfires are allowed if and only if the river is safe for swimming. - Campfires are not allowed, but the campground is open and the river is safe for swimming. - For campfires to be allowed, it is necessary and sufficient that the campground be open and that the river is not safe for swimming. - Campfires are not allowed whenever the campground is open and the river is safe for swimming.

Q5	Let p and q be the propositions: p: The project is complete. q: The team has been informed. Express each of these compound propositions as an English sentence. $\neg p$ $p \vee q$ $\neg p \wedge q$ $q \rightarrow p$ $\neg q \rightarrow \neg p$ $\neg p \rightarrow \neg q$ $p \leftrightarrow q$ $\neg q \vee (\neg p \wedge q)$
Q6	Let p, q, and r be the propositions: p: You solve the practice sheet. q: You pass the test. r: You get a good grade. Express each of these compound propositions as an English sentence. $\neg q \leftrightarrow r$ $q \rightarrow \neg r$ $p \vee q \vee r$ $(p \rightarrow \neg r) \vee (q \rightarrow \neg r)$ $(p \wedge q) \vee (\neg q \wedge r)$ $(q \wedge r) \leftrightarrow (p \oplus \neg r)$
Q7	Determine the converse, contrapositive, and inverse of each of these conditional statements (Apply De Morgan's Laws for conjunction and disjunction where necessary): - If it rains tomorrow, the outdoor concert will be canceled. - An angle is a right angle only if it measures 90 degrees. - A polygon is a triangle only if it has three sides. - You are eligible for the scholarship if your CGPA is at least 3.7 and you have completed 30 credits. - A figure is a square only if it has four equal sides and four right angles. - The machine will shut down if the temperature exceeds 100°C and the system pressure drops below the minimum threshold.
Q8	How many rows appear in a truth table for each of these compound propositions? $p \rightarrow \neg p(p \vee \neg r) \wedge (q \vee \neg s)$ $q \vee p \vee \neg s \vee \neg r \vee \neg t \vee u$ $(p \wedge r \wedge t) \leftrightarrow (q \wedge t)$
Q9	Construct truth tables for each of the following compound propositions: $p \wedge \neg p$ $p \vee \neg p$ $(p \vee \neg q) \rightarrow q$ $(p \vee q) \wedge r$ $(p \wedge q) \vee r$ $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$ $(p \vee (q \wedge r)) \rightarrow (r \vee p)$ $\neg(p \wedge q) \leftrightarrow (\neg p \vee \neg q)$ $(p \rightarrow (q \vee r)) \wedge (q \rightarrow r)$ $((p \vee q) \wedge (r \vee s)) \rightarrow (p \wedge s)$

Q10	Show that each of the following compound propositions is a tautology using the truth table. a) $p \vee \neg p$ b) $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ c) $(p \leftrightarrow q) \vee (\neg p \leftrightarrow q)$ d) $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$ e) $(\neg p \rightarrow q) \rightarrow ((\neg q \rightarrow p) \vee r)$ f) $\neg(p \wedge \neg p) \vee (q \wedge r)$
Q11	Show that each of these pairs of propositions are logically equivalent using truth tables. a) $p \rightarrow q$ and $\neg q \rightarrow \neg p$ b) $p \leftrightarrow q$ and $\neg p \leftrightarrow \neg q$ c) $(p \wedge q) \vee (\neg p \wedge \neg q)$ and $p \leftrightarrow q$ d) $p \rightarrow (q \wedge r)$ and $(p \rightarrow q) \wedge (p \rightarrow r)$ e) $\neg(p \leftrightarrow q)$ and $p \leftrightarrow \neg q$ f) $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$ g) $\neg(p \vee q)$ and $\neg p \wedge \neg q$ h) $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ i) $p \rightarrow (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$ j) $(p \vee q) \rightarrow r$ and $\neg r \rightarrow (\neg p \wedge \neg q)$
Q12	Show that, "In the Olympic Games, either the USA will win the gold medal or both Canada and Australia will win medals" and "The USA will win a medal or Canada will win a medal, and the USA will win a medal or Australia will win a medal" are logically equivalent using truth tables.
Q13	Let $D(x)$ denote "x donates to charity regularly," where the domain consists of all people in a city. Express each of these statements in English: a) $\exists x D(x)$ b) $\forall x D(x)$ c) $\exists x \neg D(x)$ d) $\forall x \neg D(x)$
Q14	Translate these statements into English, where $T(x)$ means "x is a teacher" and $R(x)$ means "x reads every day." The domain consists of all individuals in a neighborhood. a) $\forall x (T(x) \rightarrow R(x))$ b) $\exists x (T(x) \wedge R(x))$ c) $\forall x (T(x) \wedge R(x))$ d) $\exists x (T(x) \rightarrow R(x))$
Q15	Let $E(x)$ represent "x is enrolled in an art class," $B(x)$ means "x is in a beginner's course," $G(x)$ means "x has a good grade," and $S(x)$ represent "x likes studying." The domain consists of some students in an art class. Translate each of these statements into English. a) $\forall x (E(x) \rightarrow G(x))$ b) $\exists x (B(x) \wedge S(x))$ c) $\neg \forall x (B(x) \rightarrow (G(x) \vee S(x)))$
Q16	Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives. Let $F(x)$ denote "x is a family member," and $K(x)$ denote "x is kind." a) At least one family member is kind. b) All of your family members are kind. c) Not every family member is kind. d) Everyone is either a family member or is kind. e) Not everyone is a family member or there is someone who is not kind.

Q17	Let $L(x)$ represent "x knows how to play an instrument," and let $M(x)$ represent "x is a member of a band." Express each of these statements in terms of $L(x)$, $M(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at a school. - There is a student at your school who knows how to play an instrument and is a member of a band. - There is a student at your school who knows how to play an instrument but is not a member of a band. - Every student at your school either knows how to play an instrument or is a member of a band. - No student at your school knows how to play an instrument or is a member of a band.
Q18	Let $P(x)$, $Q(x)$, and $R(x)$ be the statements "x is a scientist," "x is curious," and "x likes to experiment," respectively. Express each of these statements using quantifiers, logical connectives, and $P(x)$, $Q(x)$, and $R(x)$, where the domain consists of all people. - All scientists are curious. - All curious people like to experiment. - No scientists like to experiment.
Q19	In a university, we define the following predicates: $S(x)$: x is a student. $P(x)$: x is a professor. $C(y)$: y is a course. $T(x,y)$: x takes course y. $E(x,y)$: x teaches course y. Express each of these statements using quantifiers, logical connectives. - Every student takes at least one course. - For every professor, there exists a course that they teach. - There exists a student who takes every course. - For every course, there exists a student who is enrolled in that course. - Every professor teaches at least one student in their course. - For every student, there exists a professor who teaches at least one course they are taking.
Q20	State the converse, contrapositive, and inverse of each of these conditional statements: - The ground will be wet unless it is not raining. - Having fuel is necessary for running the car. - Being sunny is sufficient for the picnic to be enjoyable. - I wake up whenever the alarm rings. - The streets get wet when it rains.
Q21	Let p and q be the propositions: p: The security system is activated. q: The alarm will sound. r: There is a breach. Write these propositions using p, q, and logical connectives (including negations). - The security system is activated, but the alarm does not sound. - The alarm sounds, and the security system is not activated. - If the security system is activated, then the alarm will sound if and only if there is a breach. - For the alarm to sound, it is necessary and sufficient that the security system is activated and there is a breach.

Q22	Let r and s be the propositions: r: I studied for the exam. s: I passed the exam. Express each of these propositions as an English sentence: a) $\neg r$ b) $r \leftrightarrow s$ c) $\neg r \oplus s$ d) $(r \leftrightarrow s) \wedge (\neg r \oplus s)$
Q23	Let p and q be the propositions p: You drive over 65 miles per hour. q: You get a speeding ticket. Write these propositions using p and q and logical connectives (including negations). a) You do not drive over 65 miles per hour. b) You drive over 65 miles per hour, but you do not get a speeding ticket. c) You will get a speeding ticket if you drive over 65 miles per hour. d) If you do not drive over 65 miles per hour, then you will not get a speeding ticket. e) Driving over 65 miles per hour is sufficient for getting a speeding ticket. f) You get a speeding ticket, but you do not drive over 65 miles per hour. g) Whenever you get a speeding ticket, you are driving over 65 miles per hour.
Q24	State the converse, contrapositive, and inverse of each of these conditional statements. a) If it snows today, I will ski tomorrow. b) I come to class whenever there is going to be a quiz. c) A positive integer is a prime only if it has no divisors other than 1 and itself.
Q25	Show that $\neg(p \wedge q) \equiv \neg p \vee \neg q$.
Q26	Show that the proposition is tautology $p \rightarrow (p \wedge (q \rightarrow p))$
Q27	Construct the truth table for $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$
Q28	Construct the truth table for $(p \oplus q) \rightarrow (p \oplus \neg q)$
Q29	Show that, "In the FIFA World Cup either Germany will reach the final or England and Argentina will reach the final" and "Germany or England will reach the final, and Germany or Argentina will reach the final" are logically equivalent.
Q30	Let $P(x)$ be the statement "x spends more than five hours every weekday in class," where the domain for x consists of all students. Express each of these quantifications in English. a) $\exists x P(x)$ b) $\forall x P(x)$ c) $\exists x \neg P(x)$ d) $\forall x \neg P(x)$
Q31	Translate these statements into English, where $C(x)$ is "x is a comedian" and $F(x)$ is "x is funny" and the domain consists of all people. a) $\forall x (C(x) \rightarrow F(x))$ b) $\forall x (C(x) \wedge F(x))$ c) $\exists x (C(x) \rightarrow F(x))$ d) $\exists x (C(x) \wedge F(x))$

Q32	Translate these statements into English, where $A(x)$: x teaches CSE230 $T(x)$: x likes STA201 $F(x)$: x has a Facebook page $C(x)$: x likes to cook The domain consists of all faculties in a university. a) $\forall x(T(x) \rightarrow F(x))$ b) $\exists x(T(x) \wedge A(x))$ c) $\neg \forall x(T(x) \rightarrow (F(x) \vee C(x)))$
Q33	Translate each of these statements into logical expressions using predicates, quantifiers, and logical connectives. a) No one is perfect. b) Not everyone is perfect. c) All your friends are perfect. d) At least one of your friends is perfect. e) Everyone is your friend and is perfect. f) Not everybody is your friend or someone is not perfect.
Q34	Let $P(x)$ be the statement "x can speak English" and let $Q(x)$ be the statement "x knows programming language Python." Express each of these sentences in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school. (a) There is a student at your school who can speak English and who knows Python. (b) There is a student at your school who can speak English but who doesn't know Python. (c) Every student at your school either can speak English or knows Python. (d) No student at your school can speak English or knows Python.
Q35	Let $P(x)$, $Q(x)$, and $R(x)$ be the statements "x is a professor," "x is ignorant," and "x is vain," respectively. Express each of these statements using quantifiers; logical connectives; and $P(x)$, $Q(x)$, and $R(x)$, where the domain consists of all people. a) No professors are ignorant. b) All ignorant people are vain. c) No professors are vain.
Q36	Let $C(x)$ means "x is a core course," $T(x,y)$ means "Student y has taken course x," and $G(y)$ means "Student y has met graduation requirements." The domain consists of all students and courses. Formulate the following statements using quantifiers: a) For every student, if they have taken all core courses, they meet the graduation requirements. b) There exists a student who has not taken every core course but still meets the graduation requirements. c) For every core course, there is at least one student who has not taken it.
Q37	Let $S(x)$ mean "x submitted a research paper," $R(x,y)$ mean "Reviewer y read x's paper," and $A(x,y)$ mean "Reviewer y approved x's paper." The domain consists of all researchers and reviewers. Express the following statements using nested quantifiers: a) Every researcher has at least one reviewer who has read and approved their paper. b) There exists a reviewer who has read every paper but has not approved at least one of them. c) For every research paper, if it was read by a reviewer, then there exists another reviewer who did not read it.

Q38	Let $P(x, y)$ mean "Person x can work on a project with Person y," $Q(x)$ means "Person x knows Python," and $R(x)$ means "Person x is a project manager." The domain consists of all people in a software development team. Express the following statements using quantifiers and logical connectives: - There exists a project manager who can work with everyone on the team who knows Python. - For every person who knows Python, there is at least one other person they can work with who is not a project manager. - For every person, if they are a project manager, there exists at least one person on the team with whom they cannot work.
Q39	Let $S(x)$ represent "x is skilled in software development," $M(x)$ represent "x has managerial experience," and $Q(x, y)$ represent "Person x qualifies for Job y." The domain consists of all applicants for the job, and there are multiple types of jobs. - Describe, using quantifiers, the condition where there exists a job such that only applicants with both software skills and managerial experience qualify. - Write the condition, using quantifiers, for a job that requires at least one applicant without software development skills but with managerial experience to qualify.
Q40	Let $A(x)$ represent "x is an applicant," $S(x)$ represent "x has strong academic scores," and $E(x)$ represent "x participates in extracurricular activities." The domain consists of all college applicants. - Express with quantifiers the requirement that all applicants participate in extracurricular activities only if they have strong academic scores, but there is at least one applicant who has strong academic scores and does not participate in extracurricular activities. - Write the condition that for an applicant to be admitted, they must either have strong academic scores or participate in extracurricular activities, but not necessarily both.

CSE230: Discrete Mathematics

Practice Sheet 2: Proofs: Direct, Indirect, Contradiction, Induction

Q1	Prove that, for all integers n , if $n^3 + 5$ is odd then n is even.
Q2	Prove the following: Suppose $a, b \in \mathbb{Z}$. If $a + b \geq 19$, then $a \geq 10$ or $b \geq 10$.
Q3	Prove the following: Suppose a, b , and c are positive real numbers. If $ab = c$ then $a \leq \sqrt{c}$ or $b \leq \sqrt{c}$.
Q4	Prove that the sum of a rational number and an irrational number is irrational.
Q5	Prove that $\sqrt{5}$ is irrational. (Same as: Prove that if x is a rational number and $x^2 = 5$, then a contradiction arises)
Q6	Prove that for any two odd integers a and b , their sum $a+b$ is even.
Q7	Prove the following: Suppose $a, b, c \in \mathbb{Z}$. If $a^2 + b^2 = c^2$, then a or b is even.
Q8	Prove the following: If a and b are positive real numbers, then $a + b \geq 2\sqrt{ab}$.
Q9	Prove by mathematical induction that if n is a positive integer then $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$
Q10	Prove by mathematical induction that if n is a positive integer then $(1 \times 2) + (2 \times 3) + (3 \times 4) + \dots + (n \times (n+1)) = \frac{n(n+1)(n+2)}{3}$
Q11	Prove by mathematical induction that $\sum_{k=1}^n k^2 = \frac{1}{6}n(n+1)(2n+1)$ for all $n \in \mathbb{N}$.
Q12	Prove by mathematical induction that if n is a positive integer then $1 + 5 + 9 + 13 + \dots + (4n-3) = \frac{n(4n-2)}{2}$
Q13	Prove by mathematical induction that, $\sum_{r=1}^n r(r+3) = \frac{1}{3}n(n+1)(n+5)$ where $n > 0$ and $n \in \mathbb{N}$.
Q14	Prove by mathematical induction that, $\sum_{r=1}^n (r-1)(r+1) = \frac{1}{6}n(n-1)(2n+5)$ where $n > 0$ and $n \in \mathbb{N}$.
Q15	Prove by mathematical induction that, $\sum_{r=2}^n r^2(r-1) = \frac{1}{12}n(n-1)(n+1)(3n+2)$ where $n > 1$ and $n \in \mathbb{N}$.

Q16	Prove by mathematical induction that, $\sum_{r=1}^n (2r-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$ where $n > 0$ and $n \in \mathbb{N}$.
Q17	Prove by mathematical induction that, $\sum_{r=1}^n r(3r-1) = n^2(n+1)$ where $n > 0$ and $n \in \mathbb{N}$.
Q18	Prove by mathematical induction that, $\sum_{r=1}^n \frac{1}{r(r+1)} = \frac{n}{n+1}$ where $n > 0$ and $n \in \mathbb{N}$.
Q19	Prove by mathematical induction that, $\sum_{r=1}^n (3^{r-1}) = \frac{3^n - 1}{2}$ where $n > 0$ and $n \in \mathbb{N}$.
Q20	Prove by mathematical induction that, $\sum_{r=1}^n r \cdot 2^r = 2 + (n-1)2^{n+1}$ where $n > 0$ and $n \in \mathbb{N}$.
Q21	Prove by mathematical induction that, $\sum_{r=1}^n (r+1) \cdot 2^{(r-1)} = n \cdot 2^n$ where $n > 0$ and $n \in \mathbb{N}$.
Q22	Prove by mathematical induction that, $-1 + 1 + 5 + 11 + \dots + \{n(n-1) - 1\} = \frac{1}{3}n(n+2)(n-2)$ where $n > 0$ and $n \in \mathbb{N}$.
Q23	Prove by mathematical induction that, $4^n + 6n + 8$ is divisible by 18 where $n > 0$ and $n \in \mathbb{N}$.
Q24	Prove by mathematical induction that, $5^{2n} + 3n - 1$ is divisible by 9 where $n > 0$ and $n \in \mathbb{N}$.
Q25	Prove by mathematical induction that, $7^n + 5$ is divisible by 6 where $n \in \mathbb{N}$.
Q26	Prove by mathematical induction that, $7^{2n-1} + 1$ is divisible by 8 where $n \in \mathbb{N}$.
Q27	Prove by mathematical induction that, $4^n + 6n - 1$ is divisible by 9 where $n \in \mathbb{N}$.
Q28	Prove by mathematical induction that, $5^n + 8n + 3$ is divisible by 4 where $n \in \mathbb{N}$.

Q29	Prove by mathematical induction that, $3^{4n} + 2^{4n+2}$ is divisible by 5 where $n \in \mathbb{N}$.
Q30	Prove by mathematical induction that, $9^n - 5^n$ is divisible by 4 where $n \in \mathbb{N}$.
Q31	Prove by mathematical induction that, $(4n + 3) \times 5^n - 3$ is divisible by 16 where $n \in \mathbb{N}$.
Q32	Prove by mathematical induction that, the sum of the cubes of any three consecutive positive integers is always divisible by 9 .
Q33	Prove by mathematical induction that, $15^n - 8^{n-2}$ is divisible by 7 where $n > 1$ and $n \in \mathbb{N}$.
Q34	Prove by mathematical induction that, $(2n + 1)7^n + 11$ is divisible by 4 where $n > 1$ and $n \in \mathbb{N}$.
Q35	Prove by mathematical induction that, $24 \times 2^{4n} + 3^{4n}$ is divisible by 5 where $n > 1$ and $n \in \mathbb{N}$.
Q36	Prove by mathematical induction that, $4 \times 7^n + 3 \times 5^n + 5$ is divisible by 12 where $n \in \mathbb{N}$.
Q37	Prove by mathematical induction that, $(2n + 1)7^n - 1$ is divisible by 4 where $n > 1$ and $n \in \mathbb{N}$.
Q38	Prove by mathematical induction that, $4^{n+1} + 5^{2n-1}$ is divisible by 21 where $n > 0$ and $n \in \mathbb{N}$.
Q39	Prove by mathematical induction that, $2^n + 6^n$ is divisible by 8 where $n > 0$ and $n \in \mathbb{N}$.
Q40	Prove by mathematical induction that, $5^{n-1} + 11^n$ is divisible by 6 where n is a positive integer.
Q41	Prove by mathematical induction that, $5^{n+1} - 4n - 5$ is divisible by 16 where $n \in \mathbb{N}$.

CSE230: Discrete Mathematics

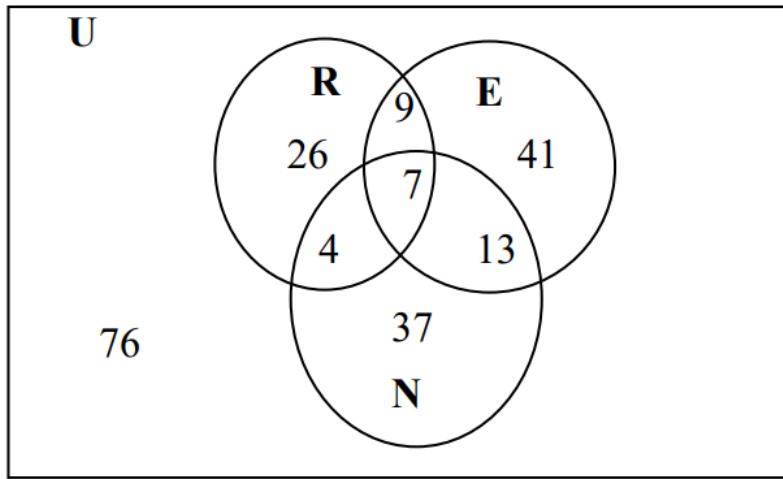
Practice Sheet 3: Sets

Q1	Determine whether each of these pairs of sets are equal. a) $\{1, 3, 3, 3, 5, 5, 5, 5, 5\}, \{5, 3, 1\}$ b) $\{\{1\}\}, \{1, \{1\}\}$ c) $\emptyset, \{\emptyset\}$
Q2	How many elements does each of these sets have where a and b are distinct elements? Here P represents the power set. a) $P(\{a, b, \{a, b\}\})$ b) $P(\{\emptyset, a, \{a\}, \{\{a\}\}\})$ c) $P(P(\emptyset))$
Q3	List all the subsets of: a) $\{1, 2, 3\}$ b) $\{\phi, \lambda, \Delta, \mu\}$ c) $\{\emptyset\}$
Q4	What is the cardinality of each of these sets? a) $\{a\}$ b) $\{\{a\}\}$ c) $\emptyset$ d) $\{\emptyset\}$ e) $\{\emptyset, \{\emptyset\}\}$ f) $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$
Q5	Find the power set of each of these sets, where p and q are distinct elements. a) $\{q\}$ b) $\{p, q\}$ c) $\{\emptyset, \{\emptyset\}\}$
Q6	Find if the following statements are True or False. a) $7 \in \{6, 7, 8, 9\}$ b) $5 \notin \{6, 7, 8, 9\}$ c) $\{2\} \not\subseteq \{1, 2\}$ d) $\emptyset \not\subseteq \{\alpha, \beta, x\}$ e) $\emptyset = \{\emptyset\}$
Q7	Let $A = \{1, 5, 31, 56, 101\}$, $B = \{22, 56, 5, 103, 87\}$, $C = \{41, 13, 7, 101, 48\}$ and $D = \{1, 3, 5, 7, \dots\}$ Give the sets resulting from: a) $A \cap B$ b) $C \cup A$ c) $C \cap D$

	d) $(A \cup B) \cup (C \cup D)$
Q8	Assume that the universal set is the set of all integers. Let, $A = \{-7, -5, -3, -1, 1, 3, 5, 7\}$ $B = \{x \in \mathbb{Z} \mid x^2 < 9\}$ $C = \{2, 3, 4, 5, 6\}$ $D = \{x \in \mathbb{Z} \mid x \leq 9\}$ In each of the following fill in the blank with most appropriate symbol from $\in, \notin, \subset, =, \neq, \subseteq$, so that resulting statement is true. $A \underline{\hspace{1cm}} D$ $3 \underline{\hspace{1cm}} B$ $9 \underline{\hspace{1cm}} D$ $\{2\} \underline{\hspace{1cm}} C^c$ $\emptyset \underline{\hspace{1cm}} D$ $A \underline{\hspace{1cm}} C$ $B \underline{\hspace{1cm}} C$ $C \underline{\hspace{1cm}} D$ $0 \underline{\hspace{1cm}} A \cap D$ $0 \underline{\hspace{1cm}} A \cup D$
Q9	Let $A = \{r, e, a, s, o, n, i, g\}$, $B = \{m, a, t, h, e, t, i, c, l\}$ and C = the set of vowels. Calculate: $A \cup B \cup C$. $A \cap B$. C^c.
Q10	Given sets A and B in a universe U, draw the Venn diagrams of each of these sets. $A \rightarrow B = \{x \in U \mid x \in A \rightarrow x \in B\}$ $A \leftrightarrow B = \{x \in U \mid x \in A \leftrightarrow x \in B\}$
Q11	Find A^3 if : $A = \{0\}$. $A = \{0, a\}$
Q12	Let $A = \{a, b, c\}$, $B = \{x, y\}$, and $C = \{0, 1\}$. Find $A \times B \times C$. $C \times B \times A$. $C \times A \times B$. $B \times B \times B$.
Q13	Given sets $A = \{1, 2, 3\}$ and $B = \{x, \{y\}\}$, find $A \times B$ and $B \times A$. Are the sets equal? What is the cardinality of the cartesian product $B \times A$?
Q14	Use a Venn diagram to illustrate the relationship $A \subseteq B$ and $B \subseteq C$.
Q15	Use a Venn diagram to illustrate the relationships $A \subset B$ and $B \subset C$.

Q1 6	Suppose that A, B, and C are sets such that $A \subseteq B$ and $B \subseteq C$. Show that $A \subseteq C$.
Q1 7	Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find a) $A \cup B$. b) $A \cap B$. c) $A - B$. d) $B - A$.
Q1 8	Let $A = \{a, b, c, d, e\}$ and $B = \{a, b, c, d, e, f, g, h\}$. Find a) $A \cup B$. b) $A \cap B$. c) $A - B$. d) $B - A$.
Q1 9	Prove the second De Morgan law that if A and B are sets, then $(A \cup B)^c = A^c \cap B^c$ by a) showing each side is a subset of the other side. b) using a membership table.
Q2 0	Use a membership table to show that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
Q2 1	Using set identities, show that $(A \cup (B \cap C))^c = (C^c \cup B^c) \cap A^c$
Q2 2	Prove the complement laws using membership table (Consider True = Universal set, False = Empty set) a) $A \cup A^c = U$. b) $A \cap A^c = \emptyset$.
Q2 3	Let A and B be two finite sets such that $n(A) = 20$, $n(B) = 28$ and $n(A \cup B) = 36$, find $n(A \cap B)$ where $n(S)$ representing the number of elements in the set S.
Q2 4	Let A, B, and C be sets. Show that a) $(A \cup B) \subseteq (A \cup B \cup C)$. b) $(A \cap B \cap C) \subseteq (A \cap B)$. c) $(A - B) - C \subseteq A - C$.
Q2 5	In a group of 200 students, 85 students are attending leadership training programs, 50 students are involved in a team-building workshop . There are 15 students who are both attending leadership training programs and participating in the team-building workshop. Determine the number of students who are involved in exactly one of the activities.

Q2
6



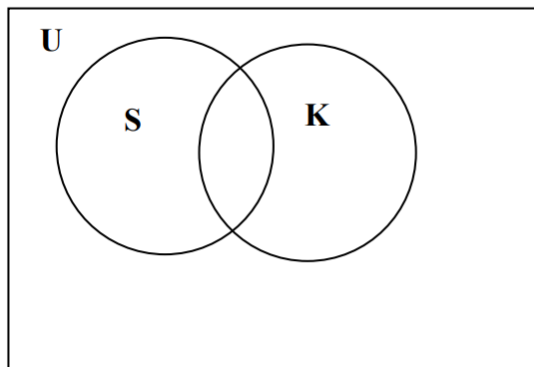
Use the diagram above to find the cardinality for each problem.

1. $n(N)$
2. $n[(R \cap E) \cup N]$
3. $n(E')$
4. $n(E \setminus R)$
5. $n(U)$
6. $n(R \cup E \cup N)$

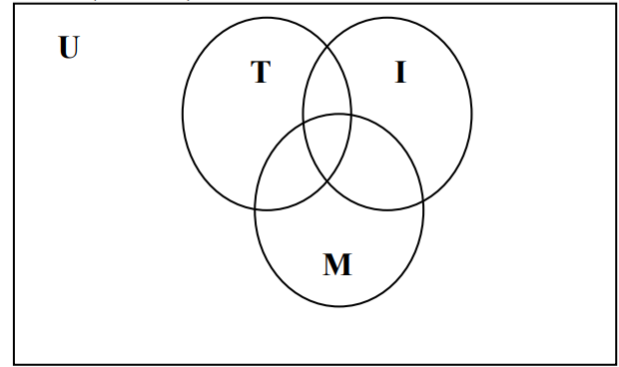
Q2
7

Illustrate each of the following by shading the Venn diagrams below.

a. $S' \cap K$

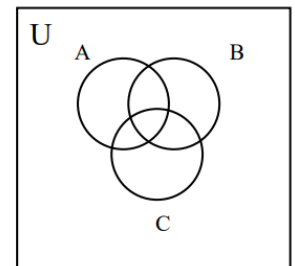
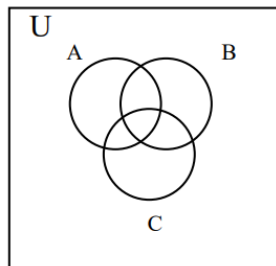
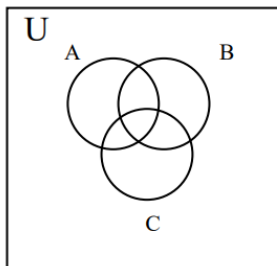


b. $(T \cup M)'$



Q2
8

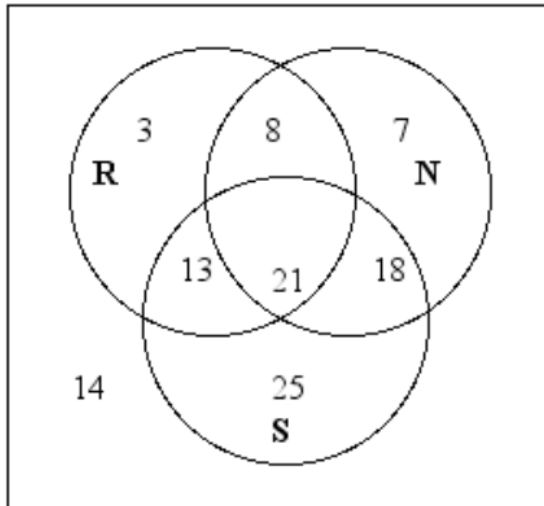
Use the Venn diagram to work the following problem in three steps $A' \cup (B \cap C)$.



Q29

A teacher surveyed her class of 43 students to find out how they prepared for their last test. She found that 24 students made flash cards, 14 studied their notes, and 27 completed the review assignment. Of the entire class of 43 students, 12 completed the review and made flash cards, 9 completed the review and studied their notes, and 7 made flash cards and studied their notes, while only 5 students completed all three of these tasks. The remaining students did not do any of these tasks. Represent this information in a Venn Diagram.

A number of deer were surveyed about activities that they enjoy. The results are summarized in the Venn diagram below:



R: enjoy running
N: enjoy nibbling
S: enjoy staring into headlights

Q30

How many deer :

- Enjoy running or staring into headlights?
- Enjoy nibbling and running?
- Enjoy exactly one of these activities?
- Enjoy staring into headlights but not running?
- Enjoy nibbling?
- Don't enjoy any of the activities?
- Enjoy any two activities only?

Q31

Draw a Venn diagram using 3 sets Q, T and P. None of these 3 sets are pairwise disjoint. Moreover, $P \cap Q \cap T \neq \phi$. How many disjoint regions are there? Indicate which regions fall under $(T \cap Q)^c - P$.

Q32

A survey of 1,200 people was conducted on three types of sports they enjoy: soccer, basketball, and baseball. Out of these, 670 enjoy soccer, 510 enjoy basketball, and 760 enjoy baseball. It was found that 320 people enjoy all three sports, while 45 people do not enjoy any of the three sports.

- How many people enjoy exactly two of these sports?
- How many people enjoy at least two of these sports?
- How many people enjoy exactly one of these sports?

CSE230: Discrete Mathematics

Practice Sheet 4: Sequence, Summation, Recurrence Solution

Q1	What is the term a_8 of the sequence $\{a_n\}$ if a_n equals a) 2^{n-1} ? b) 7 ? c) $1 + (-1)^n$? d) $-(-2)^n$?
Q2	What are the terms a_0, a_1, a_2 and a_3 of the sequence $\{a_n\}$, where a_n equals a) $(-2)^n$? b) 3 ? c) $7 + 4^n$? d) $2^n + (-2)^n$?
Q3	List the first 10 terms of each of these sequences. a) The sequence that begins with 2 and in which each successive term is 3 more than the preceding term. b) The sequence that lists each positive integer three times, in increasing order. c) The sequence that lists the odd positive integers in increasing order, listing each odd integer twice. d) The sequence whose n th term is $n! - 2^n$. e) The sequence that begins with 3, where each succeeding term is twice the preceding term. f) The sequence whose first term is 2, second term is 4, and each succeeding term is the sum of the two preceding terms. g) The sequence where the n th term is the number of letters in the English word for the index n . h) The sequence obtained by starting with 10 and obtaining each term by subtracting 3 from the previous term. i) The sequence whose n th term is the sum of the first n positive integers. j) The sequence whose n th term is $3^n - 2^n$.
Q4	Find the first five terms of the sequence defined by each of the following recurrence relations and initial conditions given below: a) $a_n = 6a_{n-1}, a_0 = 2$. b) $a_n = a_{n-1}^2, a_1 = 2$. c) $a_n = a_{n-1} + 3a_{n-2}, a_0 = 1, a_1 = 2$. d) $a_n = na_{n-1} + n^2a_{n-2}, a_0 = 1, a_1 = 1$. e) $a_n = a_{n-1} + a_{n-3}, a_0 = 1, a_1 = 2, a_2 = 0$.

Q5	Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$ if a) $a_n = 0$. b) $a_n = 1$. c) $a_n = (-4)^n$. d) $a_n = 2(-4)^n + 3$
Q6	Is the sequence $\{a_n\}$ a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$ if a) $a_n = 0$? b) $a_n = 1$? c) $a_n = 2^n$? d) $a_n = 4^n$? e) $a_n = n4^n$? f) $a_n = 2 \cdot 4^n + 3n4^n$? g) $a_n = (-4)^n$? h) $a_n = n^24^n$?
Q7	Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$ if a) $a_n = -n + 2$. b) $a_n = 5(-1)^n - n + 2$. c) $a_n = 3(-1)^n + 2^n - n + 2$. d) $a_n = 7 \cdot 2^n - n + 2$.
Q8	Find the solution to each of these recurrence relations with the given initial conditions. Use an iterative approach. a) $a_n = -a_{n-1}$, $a_0 = 5$. b) $a_n = a_{n-1} + 3$, $a_0 = 1$. c) $a_n = a_{n-1} - n$, $a_0 = 4$. d) $a_n = 2a_{n-1} - 3$, $a_0 = -1$. e) $a_n = (n + 1)a_{n-1}$, $a_0 = 2$. f) $a_n = 2na_{n-1}$, $a_0 = 3$. g) $a_n = -a_{n-1} + n - 1$, $a_0 = 7$.

Q9	A person deposits \$1000 in an account that yields 9% interest compounded annually. a) Set up a recurrence relation for the amount in the account at the end of n years. b) Find an explicit formula for the amount in the account at the end of n years. c) How much money will the account contain after 100 years?
Q10	Suppose that the number of bacteria in a colony triples every hour. a) Set up a recurrence relation for the number of bacteria after n hours have elapsed. b) If 100 bacteria are used to begin a new colony, how many bacteria will be in the colony in 10 hours?
Q11	Assume that the population of the world in 2017 was 7.6 billion and is growing at the rate of 1.12% a year. a) Set up a recurrence relation for the population of the world n years after 2017. b) Find an explicit formula for the population of the world n years after 2017. c) What will the population of the world be in 2050?
Q12	A factory makes custom sports cars at an increasing rate. In the first month only one car is made, in the second month two cars are made, and so on, with n cars made in the nth month. a) Set up a recurrence relation for the number of cars produced in the first n months by this factory. b) How many cars are produced in the first year? c) Find an explicit formula for the number of cars produced in the first n months by this factory.
Q13	An employee joined a company in 2017 with a starting salary of \$50,000 per year. Every year this employee receives a raise of \$1000 plus 5% of the salary of the previous year. a) Set up a recurrence relation for the salary of this employee n years after 2017. b) What will the salary of this employee be in 2025? c) Find an explicit formula for the salary of this employee n years after 2017. (non-homogeneous recurrence)

Q14	For each of these lists of integers, provide a simple formula or rule that generates the terms of an integer sequence that begins with the first term in each given list. Assuming that your formula or rule is correct, determine the next three terms of the sequence. a) 1, 0, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, ... b) 1, 2, 2, 3, 4, 4, 5, 6, 6, 7, 8, 8, ... c) 1, 0, 2, 0, 4, 0, 8, 0, 16, 0, ... d) 3, 6, 12, 24, 48, 96, 192, ... e) 15, 8, 1, -6, -13, -20, -27, ... f) 3, 5, 8, 12, 17, 23, 30, 38, 47, ... g) 2, 16, 54, 128, 250, 432, 686, ... h) 2, 3, 7, 25, 121, 721, 5041, 40321, ...
Q15	For each of these lists of integers, provide a simple formula or rule that generates the terms of an integer sequence that begins with the first term in each given list. Assuming that your formula or rule is correct, determine the next three terms of the sequence. a) 3, 6, 11, 18, 27, 38, 51, 66, 83, 102, ... b) 7, 11, 15, 19, 23, 27, 31, 35, 39, 43, ... c) 1, 10, 11, 100, 101, 110, 111, 1000, 1001, 1010, 1011, ... d) 1, 2, 2, 2, 3, 3, 3, 3, 3, 5, 5, 5, 5, 5, 5, 5, ... e) 0, 2, 8, 26, 80, 242, 728, 2186, 6560, 19682, ... f) 1, 3, 15, 105, 945, 10395, 135135, 2027025, 34459425, ... g) 1, 0, 0, 1, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 1, ... h) 2, 4, 16, 256, 65536, 4294967296, ...
Q16	Find the closed formula for each of the following sequences by relating them to a well known sequence. Assume the first term given is a_1. (a) 2, 5, 10, 17, 26, ... (b) 0, 2, 5, 9, 14, 20, ... (c) 8, 12, 17, 23, 30, ... (d) 1, 5, 23, 119, 719, ...
Q17	Write down the first 5 terms (starting with a_0) for each of the sequences described below. Then give either a closed formula or a recursive definition for the sequence (whichever is NOT given in the problem). (a) $a_n = \frac{1}{2}(n^2 + n)$. (b) $a_n = 2a_{n-1} - a_{n-2}$ with $a_0 = 0$ and $a_1 = 1$. (c) $a_n = na_{n-1}$ with $a_0 = 1$.

Q18	Consider the three sequences below. For each sequence, find a recursive definition. How are these sequences related? (a) 2, 4, 6, 10, 16, 26, 42, . . . (b) 5, 6, 11, 17, 28, 45, 73, . . . (c) 0, 0, 0, 0, 0, 0, 0, . . .
Q19	Write down the first few terms of the sequence given by $a_1 = 3$; $a_n = 2a_{n-1} + 4$. Then find a recursive definition for the sequence 10, 24, 52, 108, . . .
Q20	Show that $a_n = 3 \cdot 2^n + 7 \cdot 5^n$ is a solution to the recurrence relation $a_n = 7a_{n-1} - 10a_{n-2}$. What would the initial conditions needed for this to be the closed formula of the sequence?
Q21	Find a closed formula of the sequence with recursive definition $a_n = 2a_{n-1} - a_{n-2}$ and with $a_1 = 1$ and $a_2 = 2$.
Q22	Suppose you draw n lines in the plane so that every pair of lines cross (no lines are parallel) and no three lines cross at the same point. This will create some number of regions in the plane, including some unbounded regions. Call the number of regions R_n. Find a recursive formula for the number of regions created by n lines, and justify why your recursion is correct.
Q23	Consider the sequence 5, 9, 13, 17, 21, . . . with $a_1 = 5$. (a) Give a recursive definition for the sequence. (b) Give a closed formula for the nth term of the sequence. (c) Is 2013 a term in the sequence? Explain. (d) How many terms does the sequence 5, 9, 13, 17, 21, . . . , 533 have? (e) Find the sum: $5 + 9 + 13 + 17 + 21 + \cdots + 533$.
Q24	Consider the sum $4 + 11 + 18 + 25 + \cdots + 249$. How many terms (summands) are in the sum?
Q25	Consider the sequence 1, 7, 13, 19, . . . , $6n + 7$. (a) How many terms are there in the sequence? Your answer will be in terms of n. (b) What is the second-to-last term? (c) Find the sum of all the terms in the sequence, in terms of n.
Q26	Find $5 + 15 + 45 + \cdots + 5 \cdot 3^{20}$.
Q27	Solve the recurrence relation $a_n = a_{n-1} + 2^n$ with $a_0 = 5$.
Q28	Show that 4^n is a solution to the recurrence relation $a_n = 3a_{n-1} + 4a_{n-2}$.

Q29	Find the solution to the recurrence relation $a_n = 3a_{n-1} + 4a_{n-2}$ with initial terms $a_0 = 2$ and $a_1 = 3$.
Q30	Think back to the magical candy machine at your neighborhood grocery store. Suppose that the first time a quarter is put into the machine 1 Skittle comes out. The second time, 4 Skittles, the third time 16 Skittles, the fourth time 64 Skittles, etc. Find a recursive formula for how many Skittles the n th customer gets.
Q31	Let a_n be the number of $1 \times n$ tile designs you can make using 1×1 squares available in 4 colors and 1×2 dominoes available in 5 colors. (a) First, find a recurrence relation to describe the problem. Explain why the recurrence relation is correct (in the context of the problem). (b) Write out the first 6 terms of the sequence $a_1, a_2, \dots$ (c) Solve the recurrence relation.
Q32	Solve the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$. (a) What is the solution if the initial terms are $a_0 = 1$ and $a_1 = 2$? (b) What do the initial terms need to be in order for $a_9 = 30$?
Q33	The population of Utopia increases 5 percent each year. In 2000 the population was 10,000. What was the population in 1970?
Q34	Assume that a Database table has 0 records at time $n=0$. Suppose that at time n , $100n$ new records are populated into that table. The number of records increases 20 percent per minute. Write a recurrence relation and an initial condition that defines the number of records in the database table.
Q35	A wildlife reserve starts with 200 deers. The population increases by 15% annually due to reproduction and natural growth. Write a recurrence relation for the deer population at the end of n years.
Q36	A cup of coffee is initially at 90°C and is left to cool down in a room at a constant temperature of 25°C. The temperature of the coffee decreases by 10% of the difference between its current temperature and the room temperature every minute. Write a recurrence relation for the temperature of the coffee at the end of n minutes.
Q37	A new machine has been purchased for \$50,000. Its value depreciates by 12% of its current value each year. Generate a recurrence relation for the value of the machine at the end of n years.

Q38	A television channel holds a contest everyday. On day 1, the prize money is \$1000. For each day, the prize money increases by 10%, plus an additional \$5 from the CEO. Model recurrence relations for the contest. (Mention the base case). Solve the recurrence relation and find what will be the prize money on the 11th day according to the Contest.
Q 39	A person drops a tennis ball from the top of a 100 meter tall building. After each drop on the ground, the ball jumps up to the two-third of its previous height. Find a recurrence relation expressing the total distance covered by the ball before its nth drop on the ground.
Q40	Solve the following recurrence relation: $2a_{n+2} = 4a_{n+1} + 126a_n + 2 \cdot 5^n$. Here $a_0 = 0$, $a_1 = 5$.
Q41	Expand the following sums and products. That is, write them out the long way. (a) $\sum_{k=1}^{100} (3 + 4k).$ (d) $\prod_{k=2}^{100} \frac{k^2}{(k^2 - 1)}.$ (b) $\sum_{k=0}^n 2^k.$ (e) $\prod_{k=0}^n (2 + 3k).$ (c) $\sum_{k=2}^{50} \frac{1}{(k^2 - 1)}.$

CSE230: Discrete Mathematics

Practice Sheet 5: Counting

Q1	A password consists of 3 letters followed by 2 digits. How many different passwords can be created if letters and digits can repeat?
Q2	A club has 3 types of snacks and 2 types of drinks. Alternatively, they can offer 4 types of combo meals. How many choices does a customer have?
Q3	How many three-digit numbers can be formed using the digits 1, 2, 3, 4, 5 if repetition is not allowed and the number must be divisible by 5?
Q4	A committee of 3 people is to be formed from 5 men and 4 women such that there is at least one woman. How many such committees are possible?
Q5	Some companies offer 4 models of phones in 3 colors each. How many different phone options are available?
Q6	A library has 5 sections. Each section has 3 types of books. How many ways can a student choose one book if they are restricted to one section?
Q7	How many ways can you arrange the letters in the word "PERMUTATIONS" if the vowels must be together?
Q8	How many ways can you arrange the letters in the word "CALCULUS" if the first and the last character must be the same?
Q9	A 4-digit lock code is created using digits 1 to 6. How many such codes are possible if no digit can appear more than once?
Q10	In a row of 6 chairs, how many ways can 3 people sit if no two people can sit next to each other?
Q11**	How many ways can a 5-digit number be formed using the digits 0, 1, 2, 3, 4, 5 if the number must not start with 0 and repetition is not allowed? How many of those numbers will be even?
Q12	In a race of 8 people, how many ways can the top 3 positions be awarded if no ties are allowed?
Q13	In the last BRACU chess competition, there were a total of 105 participants. The rules were simple. A game of chess will be played between two of the participants randomly. The loser will be eliminated immediately. What was the total number of games played in that competition?

Q14	There are 17 points in a plane. 5 of them lie on the same straight line. No three other points lie on a single straight line. a) How many straight lines can be drawn using the points? b) How many different triangles can be drawn?
Q15*	In how many ways can the letter of the words "MATHEMATICS" be arranged maintaining the following conditions? a) Keeping the vowels together. b) Keeping the relative position of vowels and consonants unchanged. c) Starting and ending with the same letters. d) Starting with M. e) Ending with T. f) Starting with M or ending with T. g) Starting with a vowel. h) Keeping the relative position of C and S unchanged

CSE230: Discrete Mathematics

Practice Sheet 5: Pigeonhole Principle

Q16 *	An opaque container has 7 red balls, 12 yellow balls, 8 green balls, 5 blue balls and 21 black balls. What is the least number of balls you need to pick up from that container to ensure that, a) you have at least 3 balls of the same color b) you have at least 9 balls of the same color c) you have at least 2 red balls d) you have one ball of each color
Q17	The population of BRAC University students will become 300 million by 2030. There are 30 departments and each department allocates 10 million students. While forming a club, what is the minimum number of students that needs to be selected randomly so that at least 5 of the students come from the same department?
Q18	What is the minimum number of students, each of whom comes from one of the 64 districts of Bangladesh, who must be enrolled in a university to guarantee that there are at least 100 who come from the same district?
Q19	In one of the admission tests, a university decided to intake students such that at least 10 newly admitted students share the same birthday. What is the minimum number of students the university needs to admit in that intake? [Based on a real-life university]
Q20	How many numbers must be selected from the set $\{1, 3, 5, 7, 9, 11, 13, 15\}$ to guarantee that at least one pair of these numbers add up to 16?
Q21	Suppose that there are nine students in a discrete mathematics class at a small college. a) Show that the class must have at least five male students or at least five female students. b) Show that the class must have at least three male students or at least seven female students.
Q22	What is the minimum number of students required in a CSE230 class to be sure that at least six will receive the same grade if there are 13 possible grades, A+, A, A-, B+, B, B-, C+, C, C-, D+, D, D- and F?
Q23	What is the size of the smallest group of people where at least 9 people share the same birth month?
Q24 **	Show that for every integer n there is a multiple of n that has only 0s and 1s in its decimal expansion.

* symbol indicates the degree of difficulty

CSE230: Discrete Mathematics

Practice Sheet 6: Permutations and Combinations

Q1	In a class, there are 27 boys and 14 girls. The teacher wants to select 1 boy and 1 girl to represent the class for a function. In how many ways can the teacher make this selection?
Q2	(i) How many numbers are there between 99 and 1000 having 7 in the unit's place? (ii) How many numbers are there between 99 and 1000 having at least one of their digits 7?
Q3	In how many ways can 5 children be arranged in a line such that (i) two particular children of them are always together (ii) two particular children of them are never together.
Q4	How many rearrangements of the following words are possible? a. carrier, mississippi b. betterment , so that the vowels stay together. c. endeavour , so that not all the vowels are together. d. TRIANGLE so that no vowels are together. e. INTERMEDIATE such that no two vowels come together
Q5	In how many ways 3 mathematics books, 4 history books, 3 chemistry books and 2 biology books can be arranged on a shelf so that all books of the same subjects are together.
Q6	A student has to answer 10 questions, choosing at least 4 from each of Parts A and B. If there are 6 questions in Part A and 7 in Part B, in how many ways can the student choose 10 questions?
Q7	Three married couples are to be seated in a row having six seats in a cinema hall. If spouses are to be seated next to each other, in how many ways can they be seated? Find also the number of ways of their seating if all the ladies sit together.
Q8	In a small village, there are 87 families, of which 52 families have at most 2 children. In a rural development programme 20 families are to be chosen for assistance, of which at least 18 families must have at most 2 children. In how many ways can the choice be made?
Q9	A boy has 3 library tickets and 8 books of his interest in the library. Of these 8, he does not want to borrow Mathematics Part II, unless Mathematics Part I is also borrowed. In how many ways can he choose the three books to be borrowed?
Q10	There are four bus routes between A and B; and three bus routes between B and C. A man can travel round-trip in number of ways by bus from A to C via B. If he does not

	want to use a bus route more than once, in how many ways can he make a round trip?
Q11	How many committees of 4 members (2 men, 2 women) are possible if 5 men and 6 women are available?
Q12	Ten different letters of the alphabet are given. How many 5-letter words can be formed using these letters such that at least one letter is repeated?
Q13	How many signals can be sent using 6 flags of different colors, taking one or more at a time?
Q14	In an examination with three multiple-choice questions, each having 4 options, in how many ways can a student fail to get all the answers correct?
Q15	Eight chairs are numbered 1 to 8. Two women and 3 men wish to occupy one chair each. First the women choose the chairs from amongst the chairs 1 to 4 and then men select from the remaining chairs. Find the total number of possible arrangements.
Q16	In how many ways can you arrange 5 boys and 8 girls in a line so that no two boys are next to each other?
Q17	A candidate is required to answer 7 questions out of 12 questions, which are divided into two groups, each containing 6 questions. He is not permitted to attempt more than 5 questions from either group. Find the number of different ways of doing questions.
Q18	We wish to select 6 persons from 8, but if the person A is chosen, then B must be chosen. In how many ways can selections be made?
Q19	In how many ways can you divide 12 people into 3 teams A, B, C of equal members?
Q20	There are 10 people named P1 , P2 , P3 , ... P10. Out of 10 persons, 5 persons are to be arranged in a line such that in each arrangement P1 must occur whereas P4 and P5 do not occur. Find the number of such possible arrangements.
Q21	How many committees of five persons with a chairperson can be selected from 12 persons?
Q22	Find the number of permutations of n distinct things taken r together, in which 3 particular things must occur together.
Q23	18 mice were placed in two experimental groups and one control group, with all groups equally large. In how many ways can the mice be placed into three groups?
Q24	How many total words can be formed by selecting 2 vowels and 3 consonants from 4 vowels and 5 consonants?

Q25	In how many ways can a football team of 11 players be selected from 16 players? How many of them will (i) include 2 particular players? (ii) exclude 2 particular players?
Q26	A sports team of 11 students is to be constituted, choosing at least 5 from Class XI and at least 5 from Class XII. If there are 20 students in each of these classes, in how many ways can the team be constituted?
Q27	A group consists of 4 girls and 7 boys. In how many ways can a team of 5 members be selected if the team has (i) no girls (ii) at least one boy and one girl (iii) at least three girls.
Q28	A five-digit number divisible by 3 is to be formed using the digits 0,1,2,3,4,5 without repetitions. How many such numbers can be formed?
Q29	In a room, everybody shakes hands with everybody else. The total number of handshakes is 66. How many people are in the room?
Q30	How many triangles can be formed by choosing the vertices from a set of 12 points, seven of which lie on the same line?
Q31	How many ways can a team of 11 players be selected from 22 players, always including 2 specific players and excluding 4 specific players?
Q32	In a football championship, 153 matches were played. Each team played one match with every other team. How many teams participated?
Q33	A committee of 6 members is to be chosen from 10 men and 7 women such that it contains at least 3 men and 2 women. If two particular women refuse to serve on the same committee, how many such committees can be formed?
Q34	There are 3 books on Mathematics, 4 on Physics and 5 on English. How many different collections can be made such that each collection consists of : (a) One book of each subject (b) At least one book of each subject (c) At least one book of English
Q35	Five boys and five girls form a line. Find the number of ways of making the seating arrangement under the following condition: (a) Boys and girls alternate (b) No two girls sit together (c) All the girls sit together (d) All the girls are never together
Q36	There are 10 professors and 20 lecturers out of whom a committee of 2 professors and 3 lecturers is to be formed. Find (a) In how many ways committees can be formed ? (b) In how many ways a particular professor is included?

	(c) In how many ways a particular lecturer is included ? (d) In how many ways a particular lecturer is excluded ?
Q37	Using the digits 1, 2, 3, 4, 5, 6, 7, a number of 4 different digits is formed. Find (a) how many numbers are formed? (b) how many numbers are exactly divisible by 2? (c) how many numbers are exactly divisible by 25? (d) how many of these are exactly divisible by 4?
Q38	How many words (with or without dictionary meaning) can be made from the letters of the word MONDAY, assuming that no letter is repeated, if (a) 4 letters are used at a time (b) All letters are used at a time (c) All letters are used but the first is a vowel

CSE230: Discrete Mathematics

Practice Sheet 7: Number Theory

Q1	Determine whether $4 \mid 7$ and whether $4 \mid 24$
Q2	What are the quotient and remainder when a) 193 is divided by 7? b) -42 is divided by 5? c) 597 is divided by 23? d) 30201 is divided by 33? e) 0 is divided by 18? f) 3 is divided by 8?
Q3	What are the possible remainders (according to the Division Algorithm) when an integer is a) Divided by 4? b) Divided by 9?
Q4	For each of the following, find the quotient and remainder (guaranteed by the Division Algorithm) a. When 17 is divided by 3. b. When -17 is divided by 3. c. When 73 is divided by 7. d. When -73 is divided by 7. e. When 436 is divided by 27. f. When 539 is divided by 110.
Q5	Today is Tuesday. Your uncle will come after 45 days. On which day (of the week) your uncle will be coming?
Q6	Find the value of $(177 \bmod 31 \cdot 270 \bmod 31) \bmod 31$
Q7	Show that if $a \mid b$ and $b \mid a$, where a and b are integers, then $a = b$ or $a = -b$.
Q8	Prove or disprove that if $a \mid bc$, where a , b and c are positive integers and $a \neq 0$, then $a \mid b$ or $a \mid c$.
Q9	Prove that if a is an integer that is not divisible by 3, then $(a + 1)(a + 2)$ is divisible by 3.
Q10	Let m be a positive integer. Show that $a \bmod m = b \bmod m$ if $a \equiv b \pmod{m}$.
Q11	Considering a and b are integers, and $a \equiv b + 7 \pmod{19}$, show that $(a^2 - 11) \equiv (b^2 - 5b) \pmod{19}$

Q12	If we consider $50 \equiv 23 \pmod{9}$, then prove that $34 \equiv 16 \pmod{9}$ [Hint: Use the Theorem 5]
Q13	28. Find $a \text{ div } m$ and $a \text{ mod } m$ when a) $a = -991, m = 99$. b) $a = -119, m = 101$. c) $a = 10299, m = 999$. d) $a = 12346, m = 101$
Q14	Find the integer a such that a) $a \equiv 17 \pmod{29}$ and $-14 \leq a \leq 14$. b) $a \equiv -11 \pmod{21}$ and $90 \leq a \leq 110$. c) $a \equiv 24 \pmod{31}$ and $-15 \leq a \leq 15$. d) $a \equiv 99 \pmod{41}$ and $100 \leq a \leq 140$.
Q15	Use modular exponentiation to find: i) $7^{64} \text{ mod } 645$. ii) $32^{203} \text{ mod } 99$. iii) $22^{3219} \text{ mod } 243$.
Q16	i) What is the octal and hexadecimal expansion of $(11011000110\ 01011)_2$. ii) What is the hexadecimal expansion of $(125110)_{10}$. iii) What is the decimal expansion of $(7716)_8$. iv) What is the binary expansion of $(\text{BADDAD})_{16}$. v) What is the octal expansion of $(12344321)_{10}$.
Q17	Expand the decimal number $(506070)_{10}$ to the base x , where $x = ((10 \cdot 2^2)^2 + 71) \text{ mod } 23$.
Q18	If $(ab)_4$ is a base-4 integer and $(ba)_7$ is a base-7 integer, what is the largest possible value of $(a+b)_{10}$ and why? Find non-zero values for a and b such that $(ab)_4 = (ba)_7$, or prove that there are none.
Q19	Determine whether the integers in each of these sets are pairwise relatively prime : i) 14, 17, 85. ii) 21, 34, 55. iii) 25, 41, 49, 64. iv) 17, 18, 19, 23.
Q20	Consider two positive integers 5271 and 48714. Calculate the Greatest Common Divisor for the integers using Euclidean Algorithm. Also show the Least Common Multiple.
Q21	Find the greatest common divisor of the following pair of numbers using the Euclidean

	Algorithm: i) 11111, 111111. ii) 1529, 14038. iii) 750, 900. iv) 414, 662.
Q22	How many divisions are required to find $\gcd(21, 34)$ using the Euclidean algorithm ?
Q23	Find $\gcd(92928, 123552)$ and $\text{lcm}(92928, 123552)$, and verify that $\gcd(92928, 123552) \cdot \text{lcm}(92928, 123552) = 92928 \cdot 123552$
Q24	If the product of two integers is $2^2 \cdot 3^5 \cdot 5^2 \cdot 7^1 \cdot 23^2 \cdot 29^1$ and their greatest common divisor is 23345, what is their least common multiple?
Q25	Determine whether the integers in each of these sets are pairwise relatively prime. a) 11, 15, 19 b) 14, 15, 21 c) 12, 17, 31, 37 d) 7, 8, 9, 11
Q26	If p is a prime number, $p > 3$, then show that $p^2 \equiv p^4 \equiv 1 \pmod{3}$
Q27	Work out the GCD of 7105, 3185 and 2898 only using the Euclidean Algorithm . Note that, $\gcd(a, b, c) = \gcd(\gcd(a, b), c)$
Q28	Given that, the square root of a number n when divided by 11 gives a remainder of 6 where $6 < \sqrt{n} < 28$. Find the number n . Then, using modular exponentiation, calculate: $n^{125} \pmod{27}$.